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Question	Answer	Marks	Guidance
2(a)	$(F(x)=) \begin{cases} 0 & x < 0 \\ \frac{2}{9}x^2 + \frac{4}{9}x & 0 \leq x < 1 \\ \frac{1}{3}(x-2)^3 + 1 & 1 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$	M1	Attempt to integrate both parts, condone missing c , ignore domains.
		A1	Both parts correct with correct c , ignore domains.
		B1	Correct domains. Must cover all reals, may overlap at boundaries. Must have 0 for $x < 0$ and 1 for $x > 2$.
		3	
2(b)	$\frac{2}{9}m^2 + \frac{4}{9}m = \frac{1}{2}$	M1	Identifies correct interval for <i>their</i> piecewise $F(x)$ to form an equation of the form equation $F(m) = \frac{1}{2}$ OE.
	$m = \frac{1}{2}(-2 + \sqrt{13})$	A1	OE, must clearly state one answer.
		2	
2(c)	$(G(y)=) \begin{cases} 0 & y < 0 \\ \frac{2}{9}y^4 + \frac{4}{9}y^2 & 0 \leq y < 1 \\ \frac{1}{3}(y^2 - 2)^3 + 1 & 1 \leq y \leq \sqrt{2} \\ 1 & y > \sqrt{2} \end{cases}$	M1	Substitute $x = y^2$ into both parts of <i>their</i> $F(x)$, ignore domains.
		A1	Correct functions in terms of y , AEF.
		B1	Correct domain in terms of y . Domain may overlap at boundaries. Condone missing 0 for $y < 0$ but must have 1 for $y > \sqrt{2}$.
		3	

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Question	Answer	Marks	Guidance
2(d)	$G\left(\frac{1}{2}(-2 + \sqrt{13})\right) = 0.379$	M1	Evaluates $G(\text{their } m)$.
	$0.379 < 0.5$ so the median of Y is greater than the median of X .	A1	
	Alternative method for question 2(d)		
	$m_y^2 = \frac{-2 + \sqrt{13}}{2}$	M1	Expresses m_y in terms of m_x .
	$m_y^2 = m_x < 1$ and hence $m_y > m_x$.	A1	Uses the fact squaring a number between 0 and 1 makes it smaller.
	Alternative method for question 2(d)		
	Solves $\frac{2}{9}y^4 + \frac{4}{9}y^2 = \frac{1}{2}$.	M1	Identifies correct interval for <i>their</i> piecewise $G(y)$ to form equation of the form $G(y) = \frac{1}{2}$ OE.
	$m_y = 0.896 > 0.803 = m_x$	A1	0.803 may be seen in part 2(b) . A0 for $\frac{1}{2}(-2 + \sqrt{13}) < 0.896$ without $\frac{1}{2}(-2 + \sqrt{13})$ evaluated.
		2	